

Industry Project 2

Smart Building Occupancy Intelligence

Organisation

NovaSpace Technologies – Enterprise Data Platform Team

NovaSpace Technologies operates multiple office locations.

The company has been investing in smart-building technology to better understand how office spaces are being used.

Meeting rooms and collaboration spaces are equipped with non-intrusive IoT sensors that continuously capture environmental information such as:

- temperature;
- light intensity;
- sound levels;
- CO₂ concentration;
- motion detection.

The sensor data is already being collected by the organisation's Data Engineering platform and stored for analytics.

Your Role

You are a **Data Engineer in the Enterprise Data Platform Team**.

Your team is responsible for ingesting data from multiple operational systems and making reliable datasets available for reporting, analytics and downstream applications.

One of the datasets available on the platform contains readings collected from room sensors.

The Facilities Management team has approached your team with a new requirement.

They would like to know whether the available sensor data can be used to estimate **how many people are currently occupying a room**.

Your manager has asked you to build a Proof of Concept using the data already available on the platform.

Business Background

At present, Facilities Management knows that a meeting room has been booked.

However, a booked room does not necessarily mean that the room is actually being used.

For example:

- a meeting may have been cancelled;
- a room booked for eight people may only have two attendees;
- employees may use an unbooked room;
- rooms may remain empty while lights or other facilities continue operating.

The organisation would therefore like to investigate whether its existing sensor infrastructure can provide a better indication of **actual room occupancy**.

Business Requirement

Using the environmental sensor readings already available in the company's data platform, determine whether a Machine Learning model can classify the room into one of the following states:

- **0 occupants**
- **1 occupant**
- **2 occupants**
- **3 occupants**

This is a **multiclass classification problem**.

The initial objective is to build a Proof of Concept.

If successful, the prediction could eventually be consumed by downstream applications such as:

- facilities dashboards;
- meeting-room utilisation reports;
- energy-management systems;
- workplace-planning applications.

For this project, you are only responsible for developing and evaluating the classification model.

Dataset

Use the:

UCI Machine Learning Repository

Room Occupancy Estimation Dataset

Dataset ID: **864**

The dataset contains approximately:

- **10,129 observations**
- **18 sensor-related features**
- occupancy levels between **0 and 3 people**

The sensors include measurements relating to:

- Temperature
- Light
- Sound
- CO₂
- PIR motion detection

The sensor nodes transmitted readings approximately every 30 seconds.

Dataset Link

UCI Machine Learning Repository:

<https://archive.ics.uci.edu/dataset/864/room+occupancy+estimation>

Think of the Dataset as an Enterprise Data Product

In the real company environment, you would not necessarily receive a CSV file directly from the sensors.

Imagine that another part of your Data Engineering pipeline has already performed the following steps:

IoT Sensors

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Sensor Gateway



Raw Sensor Data



Enterprise Data Platform



Curated Room Sensor Table



Your Machine Learning Project

The dataset provided to you represents the **curated data layer**.

Your job begins from this point.

You should nevertheless inspect the dataset carefully rather than assuming that the data is ready for modelling.

Project Requirement

The Facilities Management team would like answers to two questions.

Question 1

Can the existing sensor measurements be used to estimate room occupancy?

Question 2

If they can, which of the classification approaches currently available to the team should be used?

You have been asked to investigate:

- Logistic Regression
- K-Nearest Neighbours
- Decision Tree

No other Machine Learning algorithms should be used.

Part 1 – Understand the Data

Begin by understanding the dataset your data platform has provided.

Determine:

- number of observations;
- number of columns;
- feature names;
- data types;
- missing values;
- duplicate observations;
- possible occupancy values;
- number of observations belonging to each occupancy class.

Identify the **target variable**.

Explain what one row of the dataset represents.

Part 2 – Data Quality Assessment

As a Data Engineer, you should not immediately begin modelling.

First assess the quality of the data.

Investigate questions such as:

- Are any sensor readings missing?
- Are there duplicate timestamps?
- Are there unusual sensor values?
- Are all sensors measured using the same scale?
- Are all occupancy classes represented equally?
- Are Date and Time useful predictors?
- Could timestamps introduce unintended patterns into the model?

Document your findings.

Part 3 – Understand the Sensors

Study the available features.

Examples include measurements from:

- Temperature sensors
- Light sensors
- Sound sensors
- CO₂ sensors
- PIR motion sensors

Investigate whether the measurements behave differently depending on room occupancy.

For example:

Temperature

Does temperature change as the number of occupants increases?

CO₂

Does CO₂ concentration appear to increase when more people are present?

Sound

Are occupied rooms generally associated with higher sound levels?

Light

Do light readings provide information about whether a room is occupied?

PIR

How strongly is detected movement associated with occupancy?

Do not merely produce visualisations.

For each important observation, explain what you believe it means.

Part 4 – Exploratory Data Analysis

Perform appropriate exploratory analysis.

Some questions you may investigate include:

- How frequently is the room empty?
- Which occupancy level occurs most often?
- How do CO₂ levels change across occupancy classes?
- How does sound vary with occupancy?

- Does PIR activity distinguish empty and occupied rooms?
- Are readings from different sensors strongly correlated?
- Are some sensors providing similar information?

You are free to investigate additional relationships you believe are useful.

Part 5 – Prepare the Data

Prepare the dataset for Machine Learning.

Your preparation should include appropriate decisions regarding:

- feature selection;
- Date and Time;
- train/test split;
- feature scaling;
- target variable;
- class distribution.

You must explain each important preprocessing decision.

Important: Feature Scaling

Pay particular attention to the scale of the variables.

For example:

Temperature may have values around:

20 – 30

while CO₂ may have values in the hundreds.

Light intensity may operate on yet another numerical scale.

This matters because **KNN calculates distance between observations**.

A variable with much larger numerical values could dominate that distance calculation.

Determine an appropriate approach and explain your reasoning.

Part 6 – Build the Models

Build the following three models.

Model 1 – Logistic Regression

Build a Logistic Regression model capable of handling the multiple occupancy classes.

Evaluate its performance.

Model 2 – K-Nearest Neighbours

Build a KNN classifier.

Experiment with different reasonable values of **K**.

Do not simply select a value randomly.

Compare the results and justify the K you finally choose.

Model 3 – Decision Tree

Build a Decision Tree classifier.

Investigate the effect of changing the depth of the tree.

Compare:

- a very small tree;
- a larger tree;
- an unrestricted tree.

Observe whether increasing tree complexity always improves performance on unseen data.

Part 7 – Model Evaluation

For each model, calculate and interpret appropriate classification metrics.

At minimum include:

- Accuracy
- Precision

- Recall
- F1 Score
- Confusion Matrix

Because this is a multiclass problem, clearly mention how Precision, Recall and F1 have been calculated.

Do not simply report the numbers.

Interpret them.

Part 8 – Confusion Matrix Analysis

The Facilities team is particularly interested in understanding **what kinds of mistakes the model makes**.

For example:

Does the model frequently confuse:

2 occupants with 3 occupants?

That may be relatively understandable.

But what if the model frequently predicts:

0 occupants when there are actually 3 occupants?

That may be much more important from a business perspective.

Examine the confusion matrix and identify the most common classification errors.

Explain what you observe.

Part 9 – Compare the Three Models

Prepare a comparison such as:

Model	Accuracy	Precision	Recall	F1
Logistic Regression				
KNN				
Decision Tree				

Then answer:

- Which model performs best?
 - Which model performs worst?
 - Does scaling affect any of the models significantly?
 - Does model complexity affect Decision Tree performance?
 - Which occupancy classes are easiest to predict?
 - Which occupancy classes are hardest to predict?
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Part 10 – Translate the Prediction Back to the Business

Assume the model predicts:

Room_Occupancy_Count = 0

What could the company potentially do with this information?

Now assume it predicts:

Room_Occupancy_Count = 3

How could that information be consumed differently?

Think about how a prediction becomes useful only when another system or business process can act on it.

Part 11 – Data Engineering Perspective

Imagine that your Proof of Concept is successful.

The Facilities team now wants predictions every few minutes for hundreds of meeting rooms.

You do **not** need to implement this system.

However, describe how you think the data might flow.

For example:

Sensors

→ **Data ingestion**

→ **Raw data**

→ **Transformation / cleaning**

→ **Feature dataset**

→ **Classification model**

→ **Prediction table**

→ **Facilities dashboard**

Explain:

- Where would the input data come from?
- What information would have to be cleaned?
- What features would be supplied to the model?
- Where could predictions be stored?
- Which downstream system could consume the predictions?

The objective is not to design a complicated architecture.

The objective is to recognise that a Machine Learning model is normally only **one component of a larger data system**.

Part 12 – Final Recommendation

Your final report should be written for the **Head of Facilities Management**, not another Machine Learning engineer.

Explain:

1. Can room occupancy be estimated from the available sensor data?

2. Which model would you recommend?

Choose from:

- Logistic Regression
- KNN
- Decision Tree

3. Why have you selected this model?
 4. How accurate is the solution?
 5. What kinds of mistakes does the model make?
 6. Which sensor measurements appear most useful?
 7. How could the organisation use the predictions?
 8. What limitations should the company know about?
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One Important Real-World Question

The source dataset was collected in a controlled environment over a limited period.

A real organisation may have:

- rooms of different sizes;
- different sensor placements;
- different office layouts;
- different lighting conditions;
- air conditioning;
- different occupancy patterns.

Therefore answer:

Would you immediately deploy this model across every office in the company?

Explain why or why not.

Suggest what additional data you would collect before moving from a Proof of Concept to production.

Deliverables

1. Jupyter Notebook

Your notebook should contain:

- Business Understanding
- Data Understanding
- Data Quality Analysis

- Exploratory Data Analysis
 - Data Preparation
 - Logistic Regression
 - KNN
 - Decision Tree
 - Model Evaluation
 - Model Comparison
 - Business Recommendation
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2. Management Presentation

Prepare a short presentation for the Facilities team.

Recommended structure:

Slide 1 – Business Requirement

Slide 2 – Available Sensor Data

Slide 3 – What We Learned From the Data

Slide 4 – Models Evaluated

Slide 5 – Model Performance

Slide 6 – Recommended Model

Slide 7 – How the Prediction Could Be Used

Slide 8 – Limitations & Next Steps

Technical Restrictions

You may use only:

- Logistic Regression
- K-Nearest Neighbours
- Decision Tree

Do not use:

- Random Forest
- Gradient Boosting

- XGBoost
- LightGBM
- Neural Networks
- Deep Learning
- AutoML

The goal is **not to discover the most sophisticated possible algorithm.**

The goal is to demonstrate that you can take data available within an organisation and turn it into a useful classification solution.